Complex factors

1

a $9ix - 54 = 9i\left(x + 6i\right)$

b $3ix - 18 = 3i\left(x + 6i\right)$

c $8ix + 32 = 8i\left(x - 4i\right)$

d $6ix + 30 = 6i\left(x - 5i\right)$

2

a $\left(x + 5i\right)\left(x - 5i\right)$

b $\left(x + 6i\right)\left(x - 6i\right)$

c $\left(x + 10i\right)\left(x - 10i\right)$

d $\left(x + 8i\right)\left(x - 8i\right)$

e $\left(x + 2i\right)\left(x - 2i\right)$

f $\left(x + 9i\right)\left(x - 9i\right)$

g $\left(x + 11i\right)\left(x - 11i\right)$

h $\left(x + 13i\right)\left(x - 13i\right)$

3

a $\left(8x + 9i\right)\left(8x - 9i\right)$

b $3\left(x + 6i\right)\left(x - 6i\right)$

c $\left(5x + 2i\right)\left(5x - 2i\right)$

d $\left(4x + 7i\right)\left(4x - 7i\right)$

e $36\left(x + 2i\right)\left(x - 2i\right)$

f $25\left(2x + 1i\right)\left(2x - 1i\right)$

g $\left(9x + 7i\right)\left(5x - 7i\right)$

h $16\left(x + 2i\right)\left(x - 2i\right)$

4

a $\left(x + 6i\right)^2$

b $\left(x + 4i\right)^2$

c $\left(x + 5i\right)^2$

d $\left(x + 2i\right)^2$

e $\left(x + 1i\right)^2$

f $\left(x + 3i\right)^2$

g $\left(x + 10i\right)^2$

h $\left(x + 7i\right)^2$

5

a $\left(x + 8i\right)\left(x + 2i\right)$

b $\left(x + 5i\right)\left(x + 3i\right)$

c $\left(x + 4i\right)\left(x + 3i\right)$

d $\left(x + 6i\right)\left(x + 5i\right)$

e $\left(x + 8i\right)\left(x + 4i\right)$

f $\left(x + 4i\right)\left(x + 10i\right)$

g $\left(x + 8i\right)\left(x + 5i\right)$

h $\left(x + 11i\right)\left(x + 7i\right)$

6

a $\left(8x + 5i\right)^2$

b $\left(7x + 4i\right)^2$

c $\left(2x + 9i\right)^2$

d $\left(4x + 5i\right)^2$

e $9\left(2x + 1i\right)^2$

f $\left(5x + 6i\right)^2$

g $\left(3x + 4i\right)^2$

h $\left(10x + 7i\right)^2$

7

a $(5x + 8i)(3x + 7i)$

b $(2x + 3i)(x + 7i)$

c $(3x + 4i)(2x - i)$

d $(4x + 9i)(2x + 6i)$

e $(8x + i)(2x - 3i)$

f $(4x + 5i)(3x - i)$

g $(2x - 9i)(x - 3i)$

h $(7x + 3i)(5x + 6i)$